

Table 1. Full MGSM multilingual-transfer results for Qwen3-4B and Gemma-3-4B-Instruct. Qwen uses 250 problems per language; Gemma uses the frozen 100-problem subset. Higher is better for every column.

Model	Language	Method	Accuracy (%) $\uparrow$	Spanish relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Qwen3-4B	Chinese	Original	71.0	0.00	1.80	1.71
		S-PID	0.0	<u>2.00</u>	0.00	0.00
		A-LQR	20.0	1.98	1.53	1.15
		H_∞ (ours)	51.0	2.00	1.87	1.52
	French	Original	60.0	0.00	1.93	1.70
		S-PID	0.0	1.92	0.00	0.00
		A-LQR	29.0	1.94	1.62	1.11
		H_∞ (ours)	48.0	1.96	1.96	1.42
	Japanese	Original	56.0	0.00	1.80	1.64
		S-PID	0.0	<u>2.00</u>	0.00	0.00
		A-LQR	32.0	1.84	1.63	1.01
		H_∞ (ours)	40.0	1.98	1.71	1.20
	Swahili	Original	15.0	0.00	1.43	1.23
		S-PID	0.0	<u>1.96</u>	0.00	0.00
		A-LQR	12.0	1.40	1.32	0.97
		H_∞ (ours)	14.0	0.92	1.44	1.06
	Telugu	Original	6.0	0.00	1.83	1.05
		S-PID	0.0	<u>2.00</u>	0.00	0.00
		A-LQR	<u>19.0</u>	1.88	<u>1.46</u>	<u>1.18</u>
		H_∞ (ours)	8.0	0.34	1.33	1.05
Gemma-3-4B-Instruct	Chinese	S-PID	<u>0.0</u>	0.00	<u>0.00</u>	<u>0.00</u>
		A-LQR	<u>0.0</u>	0.02	<u>0.00</u>	<u>0.00</u>
		H_∞ (ours)	0.0	0.04	0.00	0.00
	French	S-PID	<u>0.0</u>	0.00	<u>0.00</u>	<u>0.00</u>
		A-LQR	<u>0.0</u>	1.02	<u>0.00</u>	<u>0.00</u>
		H_∞ (ours)	0.0	1.10	0.00	0.00
	Japanese	S-PID	<u>0.0</u>	0.00	<u>0.00</u>	<u>0.00</u>
		A-LQR	<u>0.0</u>	0.00	<u>0.00</u>	<u>0.00</u>
		H_∞ (ours)	0.0	0.12	0.00	0.00
	Swahili	S-PID	<u>0.0</u>	0.00	<u>0.00</u>	<u>0.00</u>
		A-LQR	<u>0.0</u>	0.00	<u>0.00</u>	<u>0.00</u>
		H_∞ (ours)	0.0	0.20	0.00	0.00
	Telugu	S-PID	<u>0.0</u>	<u>0.00</u>	<u>0.00</u>	<u>0.00</u>
		A-LQR	<u>0.0</u>	<u>0.00</u>	<u>0.00</u>	<u>0.00</u>
		H_∞ (ours)	0.0	0.00	0.00	0.00